

The Heat Equation

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} + f(x, t)$$

$$0 \leq x \leq 1$$

$$0 \leq t \leq T$$

k : Heat conductance

f : source/sink of heat

$$u_t = k u_{xx} + f$$

Initial data: $u(x, t=0) = u_0(x)$

Boundary data: $u(x=0, t) = \alpha(t)$
 $u(x=1, t) = \beta(t)$

Simple case: $K=1, \alpha(t)=\beta(t)=0, f=0$

FT of initial data:

$$U(x,0) = \sum_{p=0}^{\infty} \hat{U}_p(0) \sin(p\pi x)$$

Solution ansatz:

$$U(x,t) = \sum_{p=0}^{\infty} \hat{U}_p(t) \sin(p\pi x)$$

$$\sum_p \hat{U}'_p(t) \sin(p\pi x) = \sum_{p=0}^{\infty} \hat{U}_p(t) (-p^2 \pi^2) \sin(p\pi x)$$

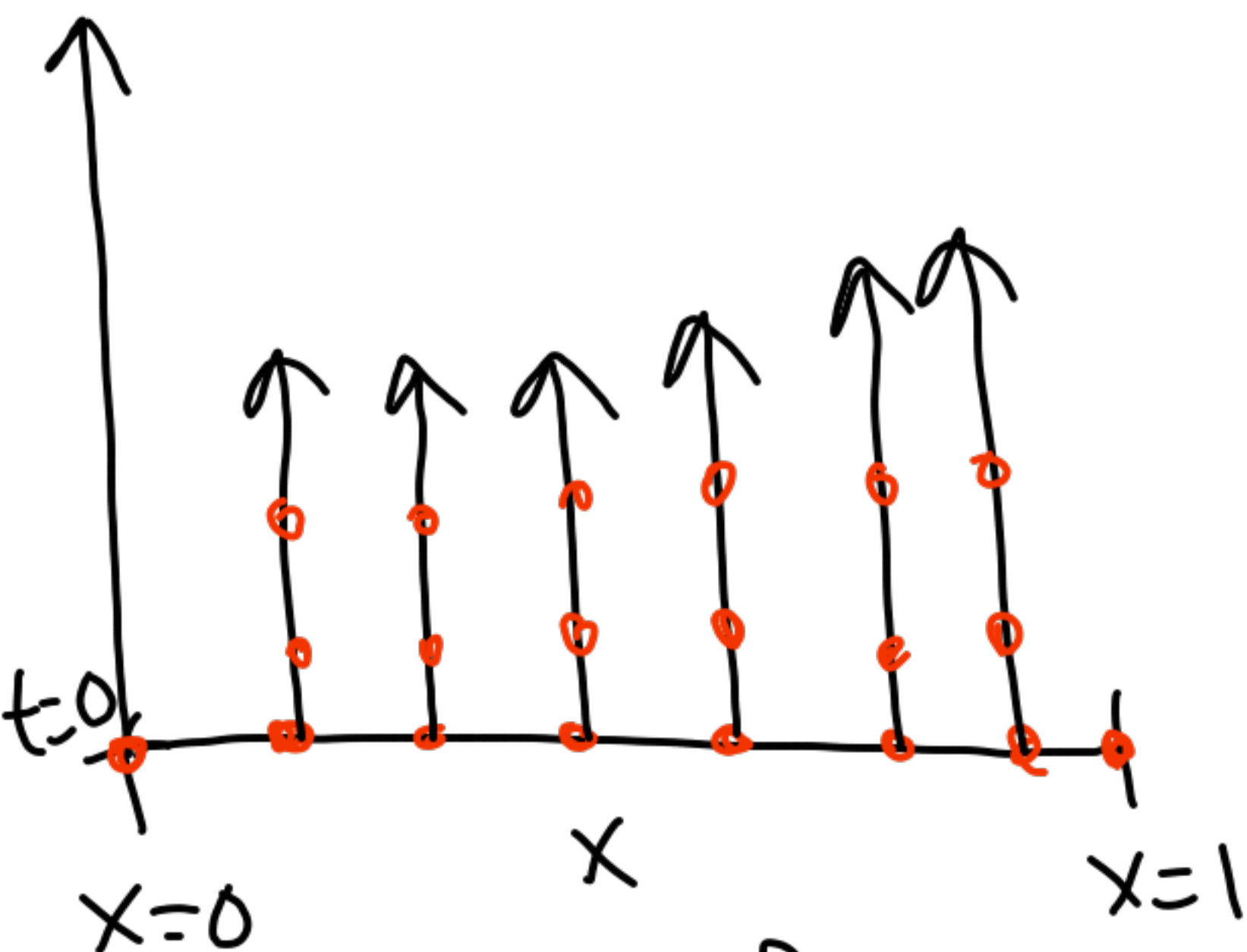
$$\hat{U}'_p(t) = -p^2 \pi^2 \hat{U}_p(t)$$

$$\hat{U}_p(t) = e^{-p^2 \pi^2 t} \hat{U}_p(0)$$

So the solution is

$$U(x,t) = \sum_{p=0}^{\infty} e^{-p^2 \pi^2 t} \hat{U}_p(0) \sin(p\pi x)$$

Discretization



Method of lines

- ① Discretize in space $\rightarrow$ ODEs
- ② Integrate in time
(RK, LM)

① Space discretization

$$x_j = jh \quad j=0, 1, \dots, m+1$$
$$h = \frac{1}{m+1}$$

$$U_j(t) \approx u(x_j, t)$$

$$u_{xx}|_{x=x_j} \approx \frac{U_{j+1}(t) - 2U_j(t) + U_{j-1}(t)}{h^2}$$

$$U_j'(t) = K \frac{U_{j+1} - 2U_j + U_{j-1}}{h^2} + f(x_j, t)$$

$$\left. \begin{array}{l} U_0(t) = \alpha(t) \quad U_{m+1}(t) = \beta(t) \end{array} \right\} j=1, 2, \dots, m$$

ODE system:

$$\begin{bmatrix} U_1' \\ \vdots \\ U_m' \end{bmatrix} = \frac{1}{h^2} \begin{bmatrix} -2 & 1 & & \\ & 1 & -2 & \\ & & \ddots & \\ & & & 1 & -2 \end{bmatrix} \begin{bmatrix} U_1 \\ \vdots \\ U_m \end{bmatrix} + \begin{bmatrix} f_1(t) \\ \vdots \\ f_m(t) \end{bmatrix}$$

$\underbrace{\hspace{15em}}_A \qquad \qquad \qquad F(t)$

$U(t)$

How does the solution of this system compare to that of the PDE?

Take $F=0$ for simplicity.

$$U(t) = e^{tA} U(0)$$

$$U'(t) = AU + F$$

$U(0)$ given (from $u_0(x)$)

Semi-discrete system

Next: apply an IVP solver

Eigenvalue decomposition:

$$AR = R\Lambda \quad R: \text{right eigenvectors}$$

Λ : eigenvalues

$$A = R\Lambda R^{-1}$$

$$\hat{U}(t) = R^{-1} U(t)$$

$$U'(t) = R\Lambda R^{-1} U(t)$$

$$(R^{-1}U)' = \Lambda R^{-1}U(t)$$

$$\hat{U}' = \Lambda \hat{U}(0)$$

$$\hat{U} = e^{t\Lambda} \hat{U}(0)$$

decay rates should
approximate $e^{-p^2\pi^2 t}$

$$\text{Eigenvalues of } A: \lambda_p = \frac{2}{h^2}(\cos(p\pi h) - 1)$$

$$\cos(p\pi h) = 1 - \frac{1}{2}p^2\pi^2 h^2 + O(h^4)$$

$$\lambda_p = -p^2\pi^2 + O(h^2)$$

we get a decay rate of

$$e^{-p^2\pi^2 t + O(p^2 h^2)}$$

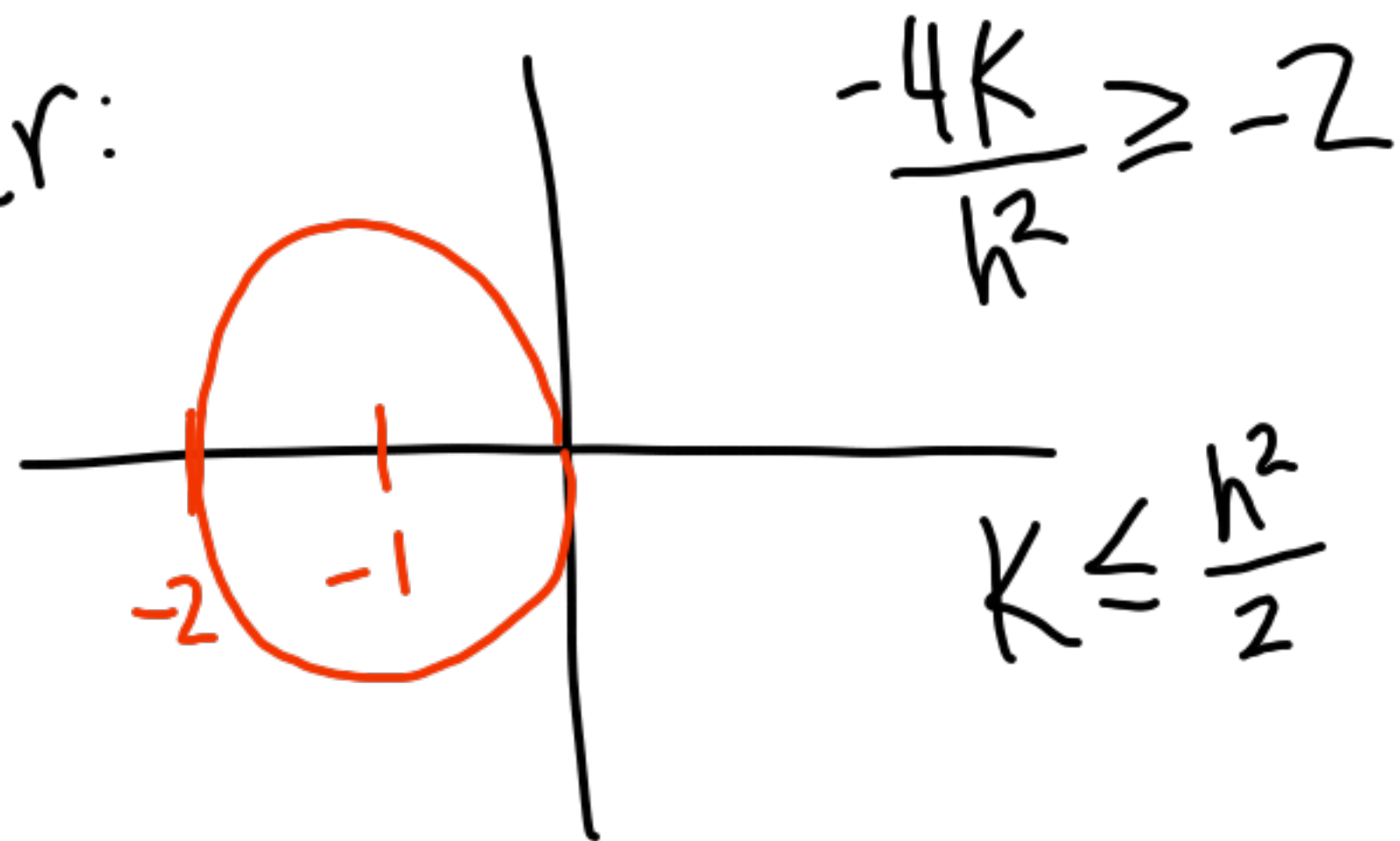
$\Rightarrow$ accurate for small values of ph .

What is the largest (magnitude)
eigenvalue?

$$-\frac{4}{h^2} \quad \text{Stiff!}$$

If we use an explicit method, we must take $K = \mathcal{O}(h^2)$ so that $K\lambda \in S$ (absolute stability)

For instance, with explicit Euler:



We should use a method that is abs. stable on the whole negative real axis, i.e.

A-stable

or at least A(0)-stable.

Then we can choose K based only on the desired accuracy.

Local Truncation Error

(Explicit Euler + Centered Difference)

$$U_j^{n+1} = U_j^n + \frac{k}{h^2} (U_{j+1}^n - 2U_j^n + U_{j-1}^n)$$

To find LTE:

$U_j^n \rightarrow u(x_j, t_n)$ and use Taylor series

$$\tau_j^n = \frac{k}{2} u_{tt}(x_j, t_n) - \frac{h^2}{12} u_{xxxx}(x_j, t_n) + \mathcal{O}(k^2, h^4)$$

1st order in time

2nd order in space